

Instructions for Students:

- If a practical cannot be performed using the website mentioned in the tutorial, you may use any other suitable demo website for practice. There is no restriction, as long as the required testing objectives are achieved.
- If you have any doubts while performing a practical, first refer to the **tutorial-wise detailed documentation PDF** available in the GitHub repository.
- If your query is still not resolved after reviewing the documentation, feel free to contact the faculty for further guidance and clarification.

For every Tutorial 1–15, students must submit:

1. Google Sheet → Test Cases

Contains all test cases designed for that tutorial.

2. Word/DOCX → Screenshot Report

Contains the **execution of exactly those same test cases**, with screenshots, actual result, Pass/Fail, and defect details where applicable.

So we should **not create unrelated activities**. The practical task, Google Sheet test cases, and screenshot report must all be connected.

SOFTWARE TESTING – TUTORIAL-WISE STUDENT TASK

Common Format for All 15 Tutorials

FILE 1: Google Sheet – Test Case

Students should use these columns:

TC ID	Test Scenario	Test Case	Preconditions	Test Steps	Test Data	Expected Result
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Example:

TC ID	Test Scenario	Test Case	Test Steps	Expected Result
TC01	Login	Verify valid login	Enter valid username/password → Click Login	User should login successfully

FILE 2: Word/DOCX – Screenshot Report

For **every test case created in the Google Sheet**, students execute it and put the evidence in the Word report.

Use:

Field	Details
TC ID	TC01

Test Case	Verify valid login
Test Data	Valid username/password
Expected Result	User should login
Actual Result	User logged in successfully
Status	PASS
Screenshot	Insert screenshot
Defect ID	N/A

Most important rule

If Google Sheet has TC01–TC10, the Word Screenshot Report must also contain TC01–TC10.

No mismatch between the two files.

TUTORIAL 1

SDLC & STLC

Website: [Automation Exercise](#)

Student Task

Study the **Login and Registration module** and identify testing activities according to SDLC/STLC.

Google Sheet Test Cases

Prepare approximately **6 test cases**:

TC ID	Test Case
TC01	Verify Login page is accessible
TC02	Verify Signup page is accessible
TC03	Verify email field is available
TC04	Verify password field is available
TC05	Verify Login functionality
TC06	Verify Logout functionality

Word Screenshot Report

Execute TC01–TC06 and insert screenshots.

Additional report:

Prepare an SDLC/STLC diagram showing where each testing activity occurs.

TUTORIAL 2

Test Plan + RTM

Website: [Automation Exercise](#)

Student Task

Prepare a **Test Plan and RTM** for Login/Registration.

Google Sheet

Prepare **8 test cases**:

1. Valid registration
2. Invalid registration
3. Existing email registration
4. Valid login
5. Invalid email
6. Invalid password
7. Blank username
8. Logout

Word Report

For each TC:

Test Case → Execution → Screenshot → Actual Result → Status

Then include:

- Test Plan
- RTM
- Test Execution Summary

TUTORIAL 3

Requirement Review / Walkthrough / Inspection

Website: [DemoBlaze](#)

Student Task

Review the requirements of the **Product and Cart module**.

Google Sheet

Prepare test cases for:

- View products

- Select product
- View product details
- Add product
- Open cart
- Verify product in cart
- Remove product

Approximately **7 test cases**.

Word Report

Execute each test case and add screenshots.

Also include a requirement-review table:

Requirement	Clear/Unclear	Reason
User can add product to cart	Clear	Testable
Website should be fast	Unclear	No response time specified

TUTORIAL 4

Functional & Non-Functional Testing

Website: [Automation Exercise](#)

Student Task

Test **Login functionality**.

Google Sheet

Prepare **10 test cases**.

Functional

- Valid login
- Invalid email
- Invalid password
- Both invalid
- Blank email
- Blank password
- Both blank

Non-functional

- Browser compatibility
- Usability
- Page layout

Word Report

Execute the test cases and insert screenshots for each executed case.

TUTORIAL 5

Defect Identification & Bug Report

Website: [SauceDemo](#)

Student Task

Test the shopping application and identify actual defects.

Google Sheet

Prepare **8–10 test cases**:

- Login
- Invalid login
- Product display
- Add to cart
- Remove from cart
- Cart
- Checkout
- Logout

Word Report

For every test case:

Execute → Screenshot → Pass/Fail

If an actual defect is found:

BUG-001

Include:

- Bug title
- Severity
- Priority
- Steps to reproduce
- Expected result
- Actual result
- Screenshot

Important: Students should report actual observed defects, not invent them.

TUTORIAL 6

Manual Test Execution

Website: [DemoBlaze](#)

Student Task

Test the complete shopping workflow.

Open Website



Select Product



Product Details



Add to Cart



Cart



Place Order



Customer Details



Submit Order

Google Sheet

Prepare **10–12 test cases**.

Word Report

Execute **the same 10–12 test cases**.

For each:

TC ID + Screenshot + Actual Result + Status

At the end:

- Total Test Cases
- Passed
- Failed

- Blocked
- Pass %

TUTORIAL 7

Boundary Value Analysis & Equivalence Partitioning

Website: [Selenium Web Form](#)

Student Task

Use a given validation requirement:

Age must be between 18 and 60.

Google Sheet

Prepare BVA test cases:

TC ID	Input	Expected
BVA01	17	Invalid
BVA02	18	Valid
BVA03	19	Valid
BVA04	59	Valid
BVA05	60	Valid
BVA06	61	Invalid

Prepare EP cases:

TC ID	Equivalence Class	Input
EP01	Below 18	15
EP02	18–60	30
EP03	Above 60	65

Word Report

For each test case, show:

Input → Execution → Screenshot → Actual Result → Pass/Fail

TUTORIAL 8

Bugzilla & TestLink

Website/Application: Use the application tested in previous tutorials.

Tools

[Bugzilla](#)

[TestLink](#)

Google Sheet

Prepare test cases such as:

- Create test case
- Execute test case
- Mark Pass
- Mark Fail
- Create defect
- Assign severity
- Assign priority
- Retest defect

Word Report

Students should provide screenshots of:

1. TestLink project
2. Test suite
3. Test case
4. Test execution
5. Bugzilla defect
6. Defect status
7. Final execution result

TUTORIAL 9

Black Box, White Box, Integration & System Testing

Website: [OrangeHRM Demo](#)

Google Sheet

Prepare test cases categorized as:

Black Box

- Valid login
- Invalid login
- Blank login

Integration

- Login → Dashboard
- Dashboard → Employee Management

System

- Complete workflow

Word Report

Each test case should have:

Testing Type → TC ID → Execution → Screenshot → Result

For White Box testing, students can use a small Python program supplied by you and document statement/branch/path coverage separately.

TUTORIAL 10

Automation & Regression Testing

Website: [SauceDemo](#)

Google Sheet

Prepare regression test cases:

TC ID	Test
REG01	Login
REG02	Product display
REG03	Add to cart
REG04	Remove from cart
REG05	Checkout
REG06	Logout

Word Report

Students execute these test cases and provide screenshots.

Then add:

Regression Testing Result

TC ID	Previous Result	Current Result	Status
REG01	Pass	Pass	Pass

REG02	Pass	Pass	Pass
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TUTORIAL 11

Python + Selenium Navigation & Multi-page Workflow

Website: [Automation Exercise](#)

Google Sheet

Prepare automation test cases:

- Open website
- Products
- Search product
- Select product
- Add to cart
- Open cart

Word Report

For each test case include:

TC ID

Python code screenshot

Browser execution screenshot

Actual result

Pass/Fail

TUTORIAL 12

AI-Assisted Software Testing

Website: [Automation Exercise](#)

Student Task

Select **Login**.

Ask Generative AI to generate test cases.

Students must:

1. Generate test cases using AI.
2. Review them.
3. Correct them.
4. Put final test cases in Google Sheet.

5. Execute them.
6. Capture screenshots.
7. Compare AI-generated expectations with actual behavior.

Google Sheet

Include an additional column:

AI Generated / Student Modified

Word Report

Show:

AI Prompt → AI Output → Final Test Case → Execution Screenshot → Result

This makes Tutorial 12 clearly different from the other tutorials.

TUTORIAL 13

Python + Selenium Environment Configuration

Website: [Selenium Web Form](#)

This is mainly a **setup practical**, so students don't need 15 test cases.

Google Sheet

Prepare **5 verification test cases**:

TC ID	Test
SET01	Verify Python installation
SET02	Verify Selenium installation
SET03	Verify browser launches
SET04	Verify Selenium Web Form opens
SET05	Verify basic Selenium script executes

Word Report

Screenshots:

1. Python version
2. Selenium installation
3. IDE/project
4. Code
5. Browser
6. Successful execution

TUTORIAL 14

Selenium Web Elements

Website: [Selenium Web Form](#)

Google Sheet

Prepare test cases for:

TC ID	Web Element
SEL01	Text box
SEL02	Password
SEL03	Text area
SEL04	Dropdown
SEL05	Checkbox
SEL06	Radio button
SEL07	Date
SEL08	Submit
SEL09	Result verification

Word Report

For each TC:

Code Screenshot → Browser Screenshot → Result

TUTORIAL 15

Login, Authentication, Assertion & Exception Handling

Website: [SauceDemo](#)

Google Sheet

Prepare **10 test cases**:

TC ID	Test Case
LOGIN01	Valid login
LOGIN02	Invalid username
LOGIN03	Invalid password
LOGIN04	Blank username
LOGIN05	Blank password
LOGIN06	Verify successful authentication

LOGIN07	Verify page title using assertion
LOGIN08	Verify logout
LOGIN09	Handle missing element exception
LOGIN10	Verify error message

Word Report

Students must show screenshots of:

1. Valid login
2. Invalid login
3. Authentication result
4. Assertion result
5. Exception handling
6. Logout

Final Submission Structure for Students

Every Tutorial = 2 Files

File 1: Google Sheet

T01_TestCases_fullenrollment no.excel

File 2: Word Screenshot Report

T01_ScreenshotReport_fullenrollment no.docx

Then:

T01 → Test Case Sheet + Screenshot Report

T02 → Test Case Sheet + Screenshot Report

T03 → Test Case Sheet + Screenshot Report

...

T15 → Test Case Sheet + Screenshot Report

The most important rule

Google Sheet = WHAT YOU PLAN TO TEST

Word Report = PROOF THAT YOU TESTED IT

And the **TC IDs must remain identical in both files.**

For example:

Google Sheet → LOGIN01

Word Report → LOGIN01 → Screenshot → Actual Result → PASS